tile mapping, you'll see that you can combine more traditional web mapping techniques with the advanced functionality provided by TopoJSON and D3's geo functions to make incredibly sophisticated web maps.

7.5 Tile mapping with d3.geo.tile

So far you've made choropleth maps, some of which are simple and others, like the satellite projection or the globe, rather exotic. But none of your maps have terrain, or satellite imagery. That kind of data—*raster* or *image* data—isn't nearly as lightweight as vector data. Think about the size of a picture you take with the camera on your phone, and imagine how large an image must be if you want to give your user the ability to zoom in to any street in the world.

To get around the problem of these massive images, web mapping uses tiles to display satellite and terrain data. A high-resolution satellite image of a city, for instance, would be cut into 256- by 256-px tiles at as many zoom levels as are appropriate and stored on a server in directories indicating the zoom and position of those tiles. It sounds like it might be a lot of work to make tiles, but fortunately, you don't have to, because companies like Mapbox (mapbox.com) provide you with tiles and the tools, like TileMill, to customize them. (Both free and commercial versions are available, depending on how many visitors your site receives.)

If you open up tile.js and take a look at it, you'll see that it's a small file. That's because geotiles are simple. Each tile is a raster image (typically a PNG) that represents one square of the earth somewhere, as you see in figure 7.20. Its filename indicates the



Figure 7.20 Your first tiled map, using pregenerated tiles from Mapbox